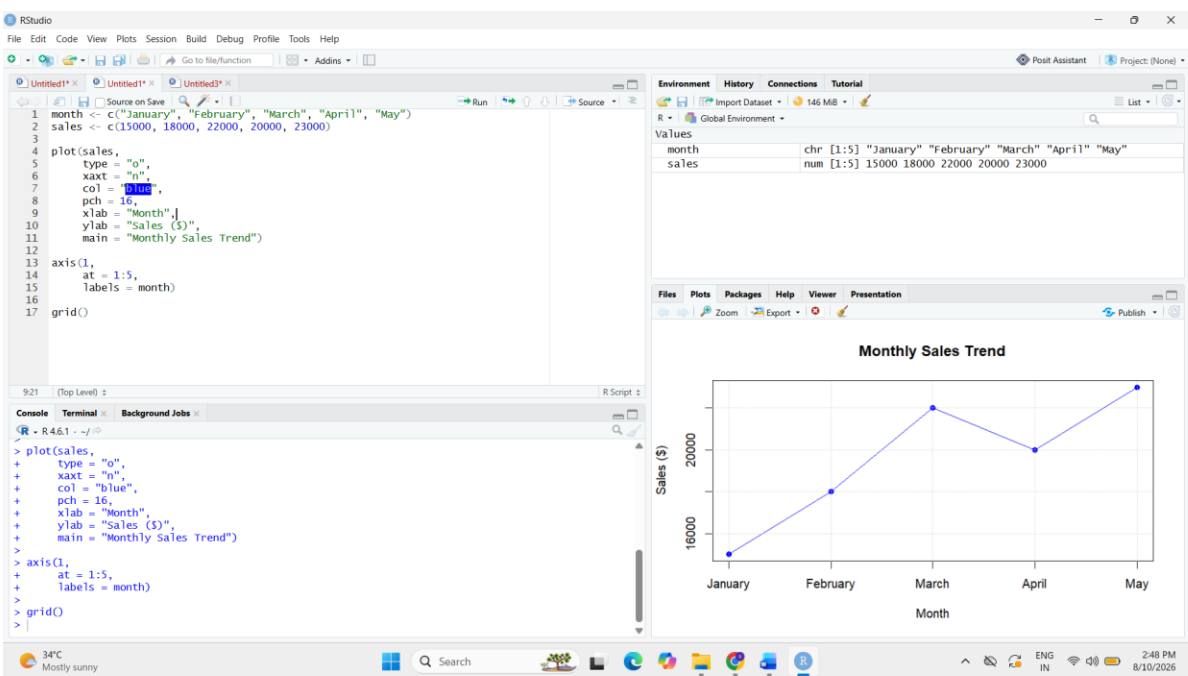
EXPERIMENT- 18

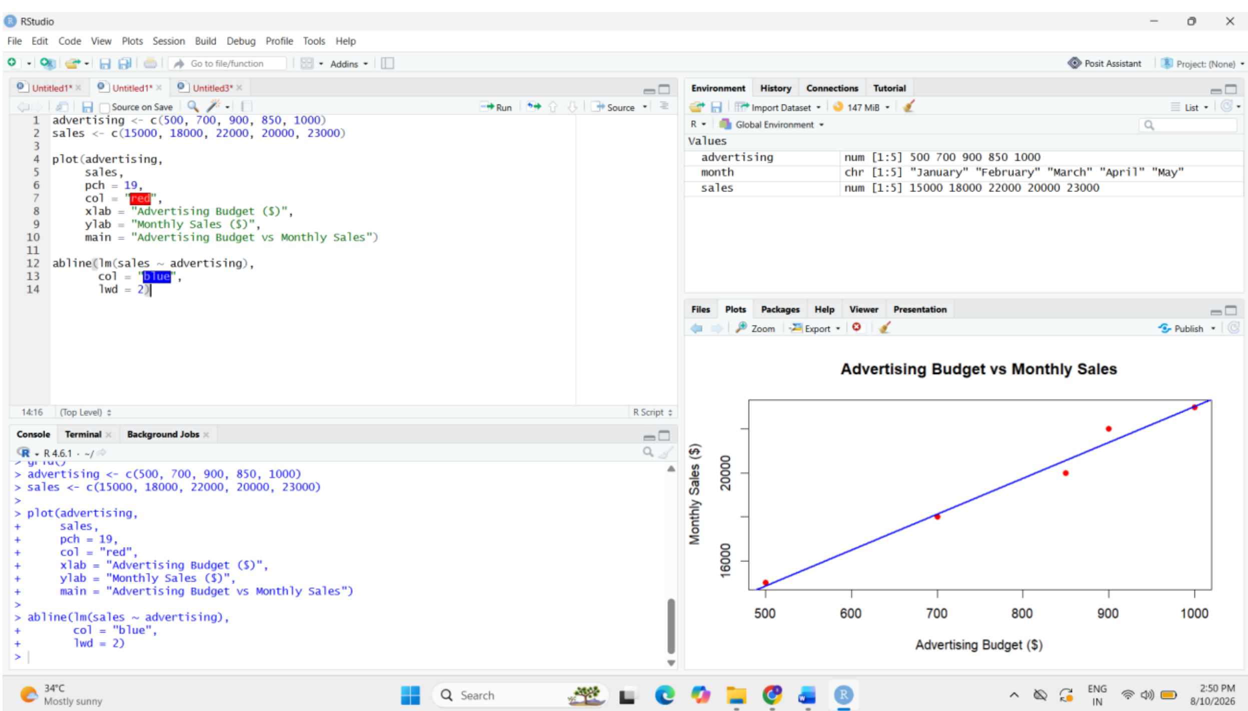
```
5.month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)
plot(sales,
    type = "o",
    xaxt = "n",
    col = "blue",
    pch = 16,
    xlab = "Month",
    ylab = "Sales ($)",
    main = "Monthly Sales Trend")
axis(1,
    at = 1:5,
    labels = month)
grid()
```



```
6.advertising <- c(500, 700, 900, 850, 1000)
sales <- c(15000, 18000, 22000, 20000, 23000)

plot(advertising,
    sales,
    pch = 19,
    col = "red",
    xlab = "Advertising Budget ($)",
    ylab = "Monthly Sales ($)",
    main = "Advertising Budget vs Monthly Sales")

abline(lm(sales ~ advertising),
      col = "blue",
      lwd = 2)
```



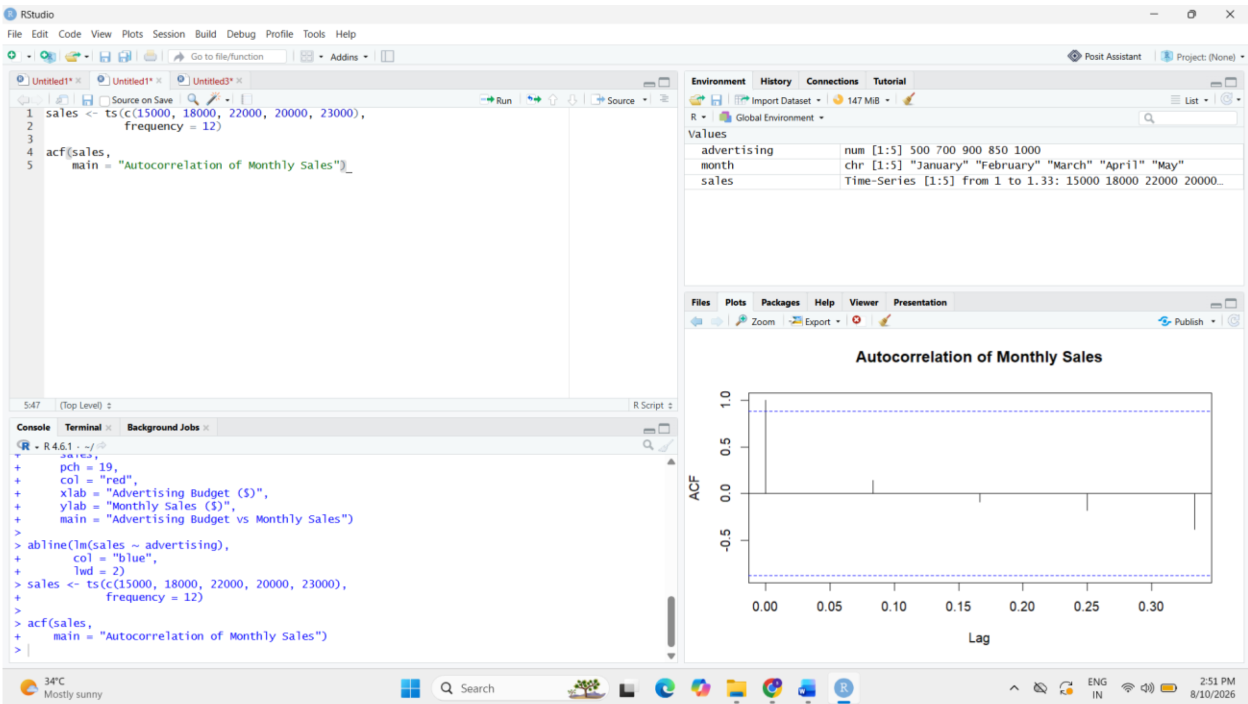
8.

sales <- ts(c(15000, 18000, 22000, 20000, 23000),

frequency = 12)

acf(sales,

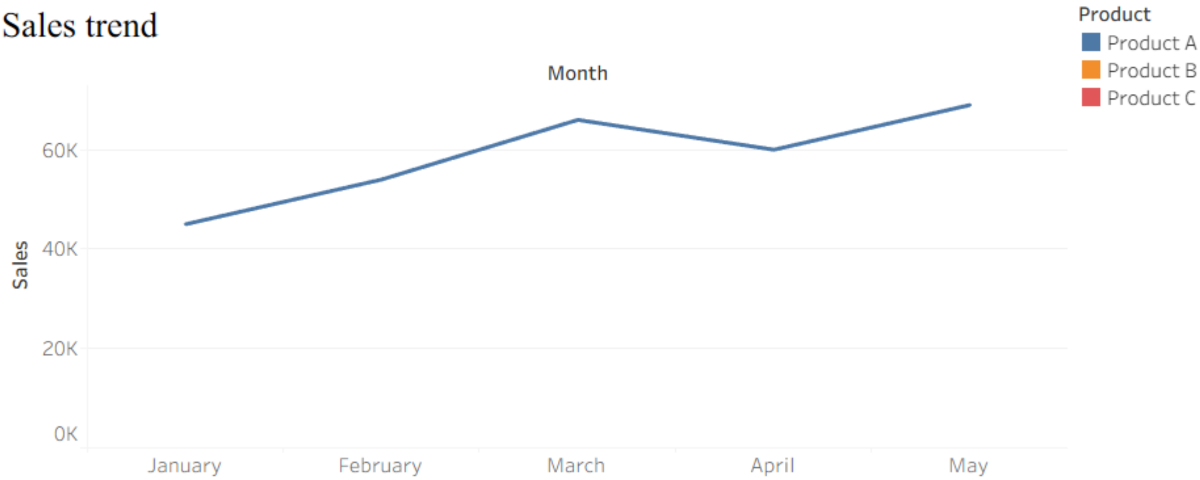
main = "Autocorrelation of Monthly Sales")



4.

Time Series Analysis

Sales trend



The distribution of sales across products